

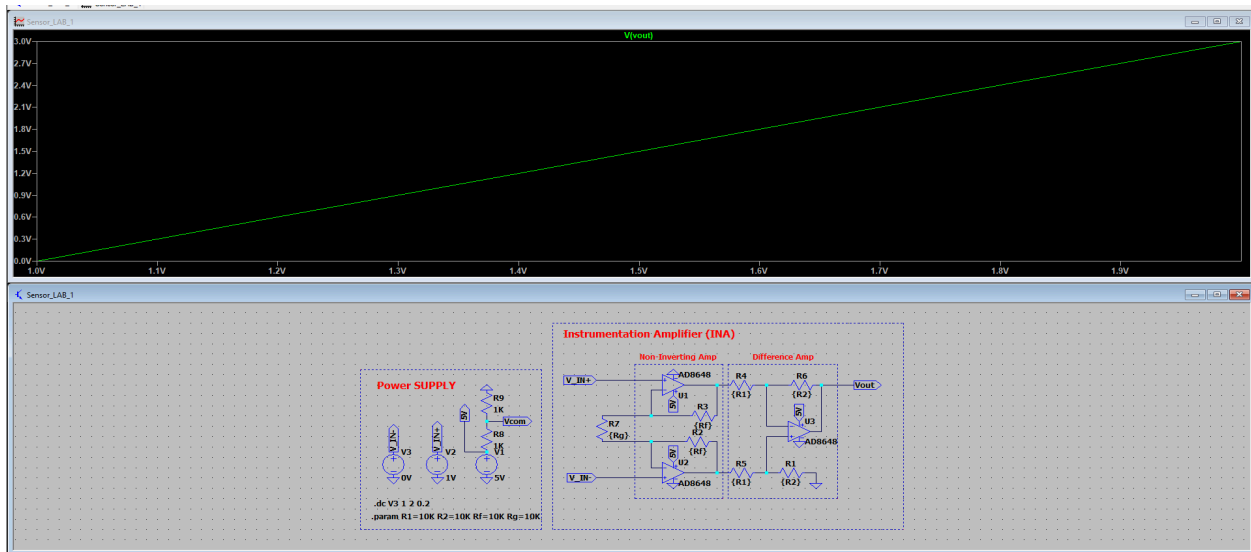
Sensors Laboratory Experiment -1

Roll No. - ES24f1100064

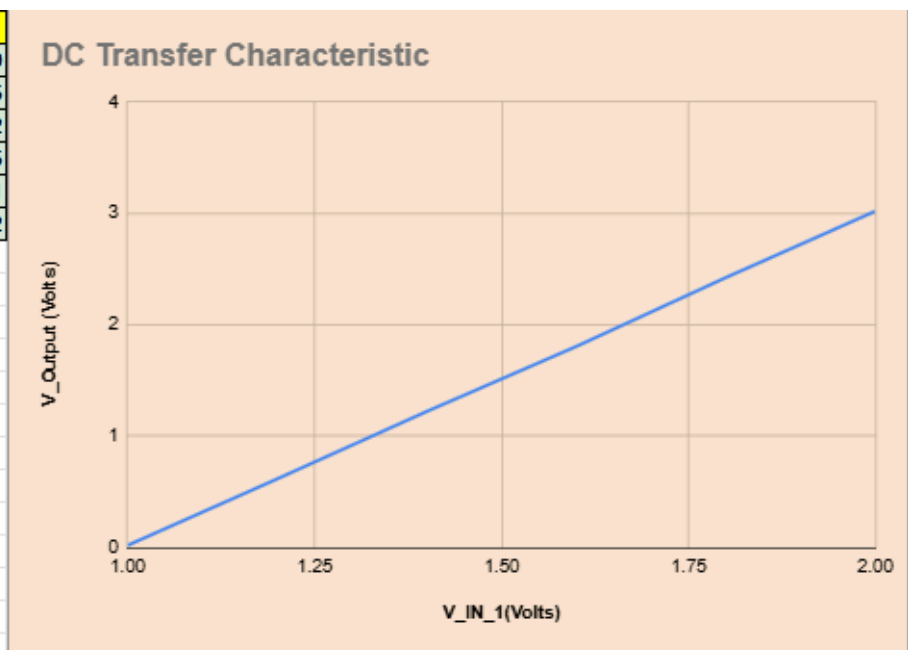
Name : Arpit Katiyar

Design and Analysis of an Instrumentation Amplifier (INA)

1. DC Transfer characteristic of simulation and real circuit agree



V_IN_1	V_output
1	0.019
1.2	0.615
1.4	1.22
1.6	1.805
1.8	2.421
2	3.02



2. When $A_d = 3$, $V_{out} = A_d * V_d + A_c * V_c$

Since $V_d = 1.5 - 1.5 = 0$,

$V_{out} = A_c * V_c$

$V_{out} = 0.019V$, $V_c = (1.5 + 1.5)/2 = 1.5V$

$A_c = 0.019/1.5 = 0.0127$

$CMRR = 20 \log (A_d/A_c) = 20 \log(3/0.0127) = 47.47 \text{ dB}$

3. Effect of variation in frequency: Gain decreases if frequency becomes very high

Frequency(Hz)	Vd	Vout	Gain (A_d)
10	0.5	0.506	1.012
100	0.5	0.528	1.056
1000	0.5	0.522	1.044
10000	0.5	0.471	0.942
20000	0.5	0.303	0.606

4. $A_d = 20$. The output voltage does not always match with the calculated voltage. It only matches for a few values. Two sets of readings were taken to get some clarity.

Units : Volts	Steps of 50 mV		
V1	V2	Vd	Vout
0.5	0.5	0	0.147
0.55	0.5	0.05	1.015
0.6	0.5	0.1	1.112
0.65	0.5	0.15	1.204
0.7	0.5	0.2	1.297
V1	V2	Vd	Vout
1	1	0	0.145
1.05	1	0.05	1.138
1.1	1	0.1	2.037
1.15	1	0.15	2.133
1.2	1	0.2	2.224

At $V_1 = V_2 = 1.5V$, $V_{out} = 0.153V$,
 $A_c = 0.153/1.5 = 0.102$
 $CMRR = 20 \log (20/0.102) = 45.84 \text{ dB}$

5. When V_{ref} was set to 2.5V and V_{IN_1} and V_{IN_2} were both set to 0, V_{out} was almost equal to V_{ref} both in simulation and in real circuit. This agrees with the derived expression for V_{out} when V_{ref} is not zero.

$$V_{out} = (1 + 2R/R_g) * V_d + V_{ref}$$

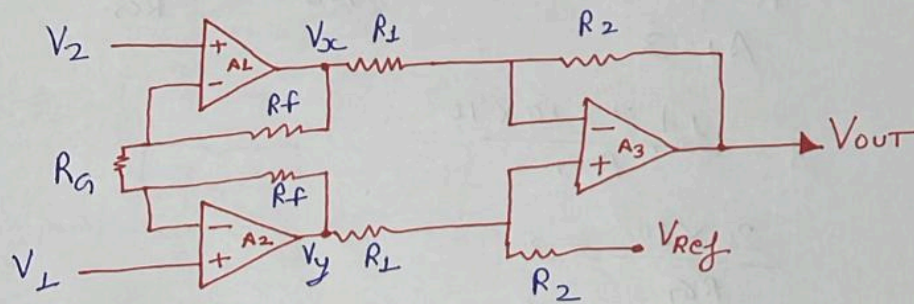
Elaborate Details:

Experiment -1:

Arpit K.
24f1100064

Design & Analysis of an Instrumentation (INA)

Task -1



Assume $V_{ref} = 0$

$$V_x = \left(1 + \frac{R_f}{R_G}\right) V_2 - \frac{R_f}{R_G} V_1$$

$$V_y = \left(1 + \frac{R_f}{R_G}\right) V_1 - \frac{R_f}{R_G} V_2$$

$$V_{OUT} = \frac{R_2}{R_1} (V_y - V_x)$$

$$= \frac{R_2}{R_1} \left[\left(1 + \frac{R_f}{R_G}\right) V_1 - \frac{R_f}{R_G} V_2 - \left(1 + \frac{R_f}{R_G}\right) V_2 + \frac{R_f}{R_G} V_1 \right]$$

$$= \frac{R_2}{R_1} \left[\left(1 + \frac{2R_f}{R_G}\right) V_1 - \left(1 + \frac{2R_f}{R_G}\right) V_2 \right]$$

$$V_{OUT} = \frac{R_2}{R_1} \left(1 + \frac{2R_f}{R_G}\right) [V_1 - V_2]$$

Gain A_d , if $R_f = R_1 = R_2 = R$ & $V_{ref} = 0$

$$A_d = 1 + \frac{2R}{R_G}$$

$$V_{OUT} = 1 + \frac{2R}{R_G} [V_1 - V_2]$$

Task -2

$$R = 10k\Omega$$

$$A_d = 3$$

$$A_d = 1 + \frac{2R}{R_G}$$

$$3 = \frac{1 + 2 \times 10 \times 10^3}{R_G}$$

$$\frac{20 \times 10^3}{R_G} = 2$$

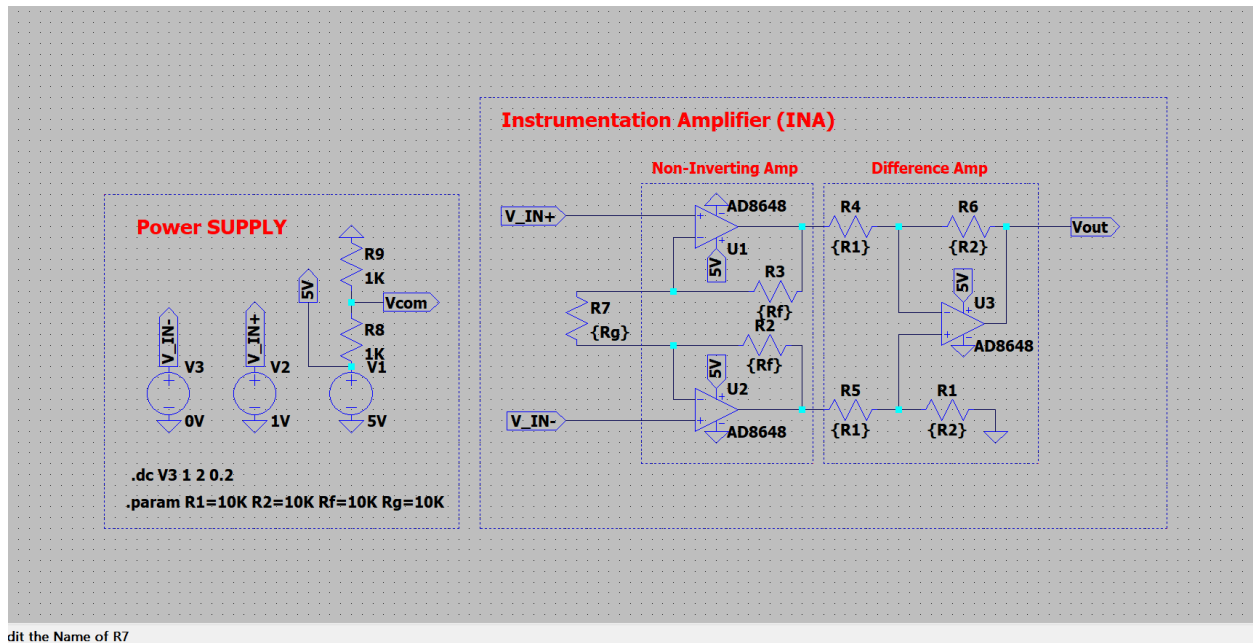
∴

$$R_G = 10k\Omega$$

Task – 3:

LTSpice Simulation

$V_{ref} = 0$, $R_1 = R_2 = R_f = R_g = 10K$ Ohms



Task 3a: DC Transfer Characteristic

V_{out} on Y-Axis, V_{IN_1} on X-axis

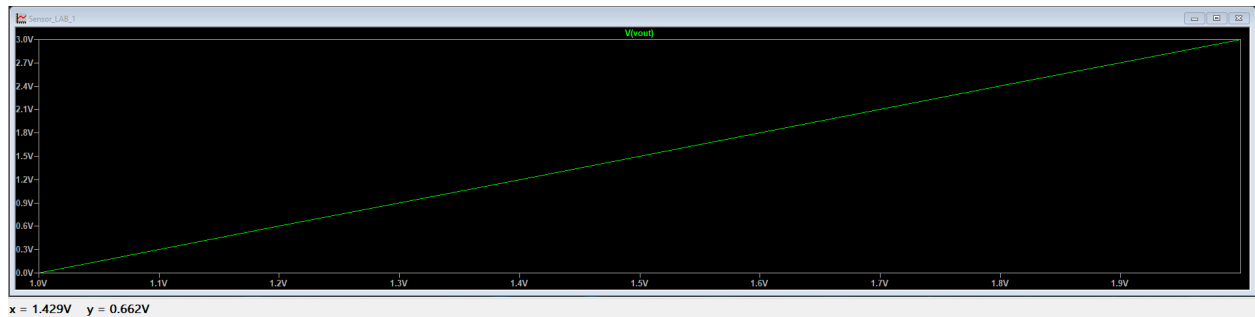
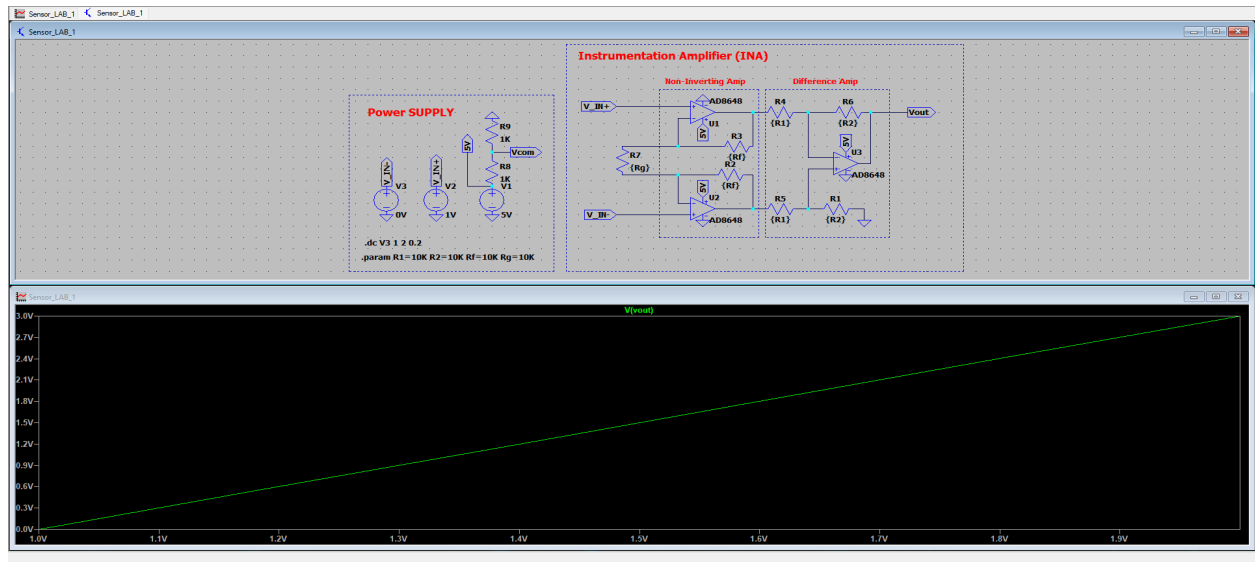
V_{IN_2} is set as 1V.

V_{IN_1} is varied from 1V to 2V in steps of 0.2V

$$V_d = V_{IN_1} - V_{IN_2}$$

From the graph, it is clear that the gain $A_d = 3$, as we calculated earlier.

That is, $V_{out} = 3 * V_d$



Task 4:

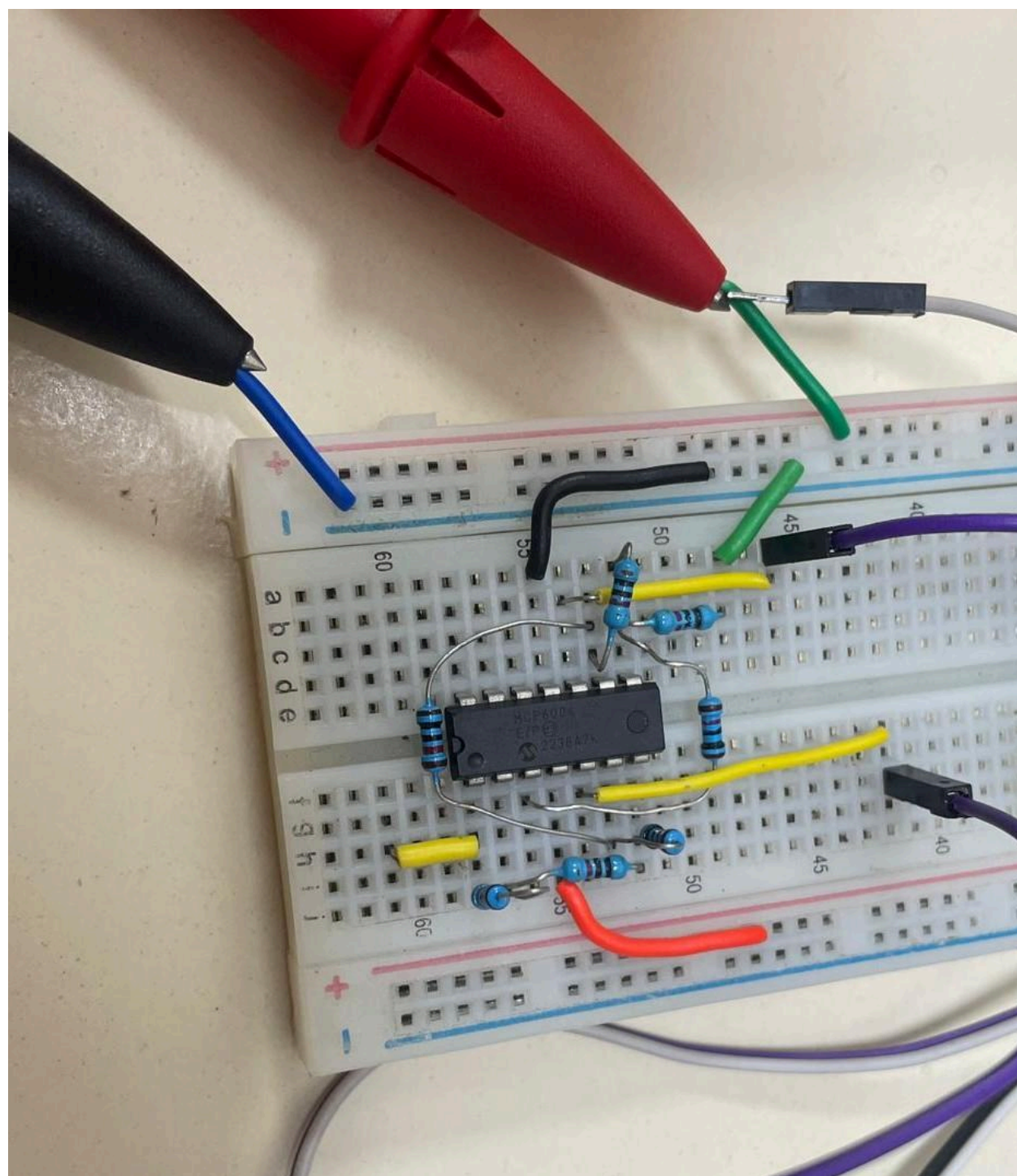
Circuit on the breadboard

One MCP6004 IC used for the three opamps.

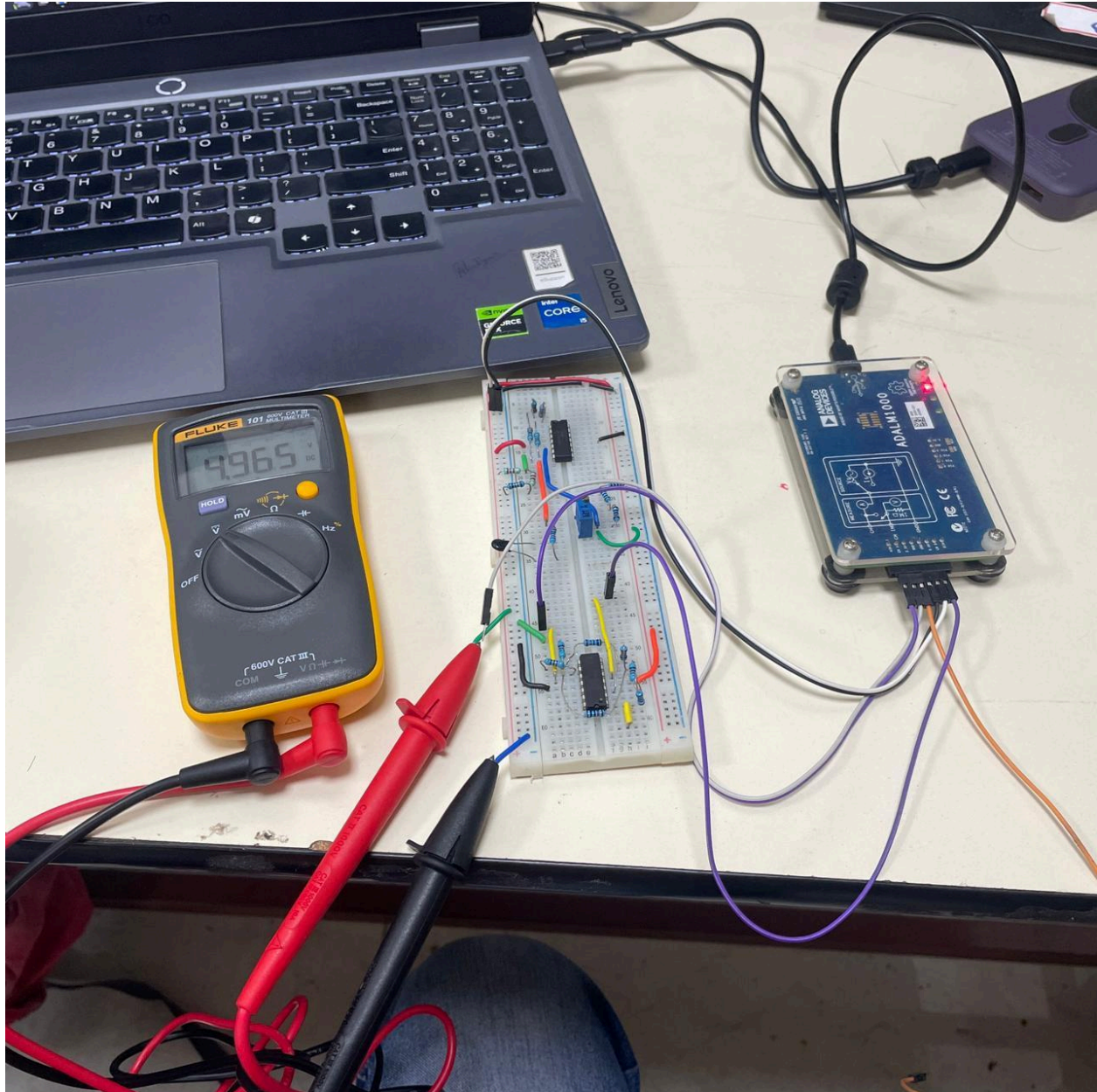
Seven 10 K Ohm resistors used for R1, R2, Rf and Rg

Channel A and Channel B of ADALM used as Voltage Sources for
V_IN_1 and V_IN_2

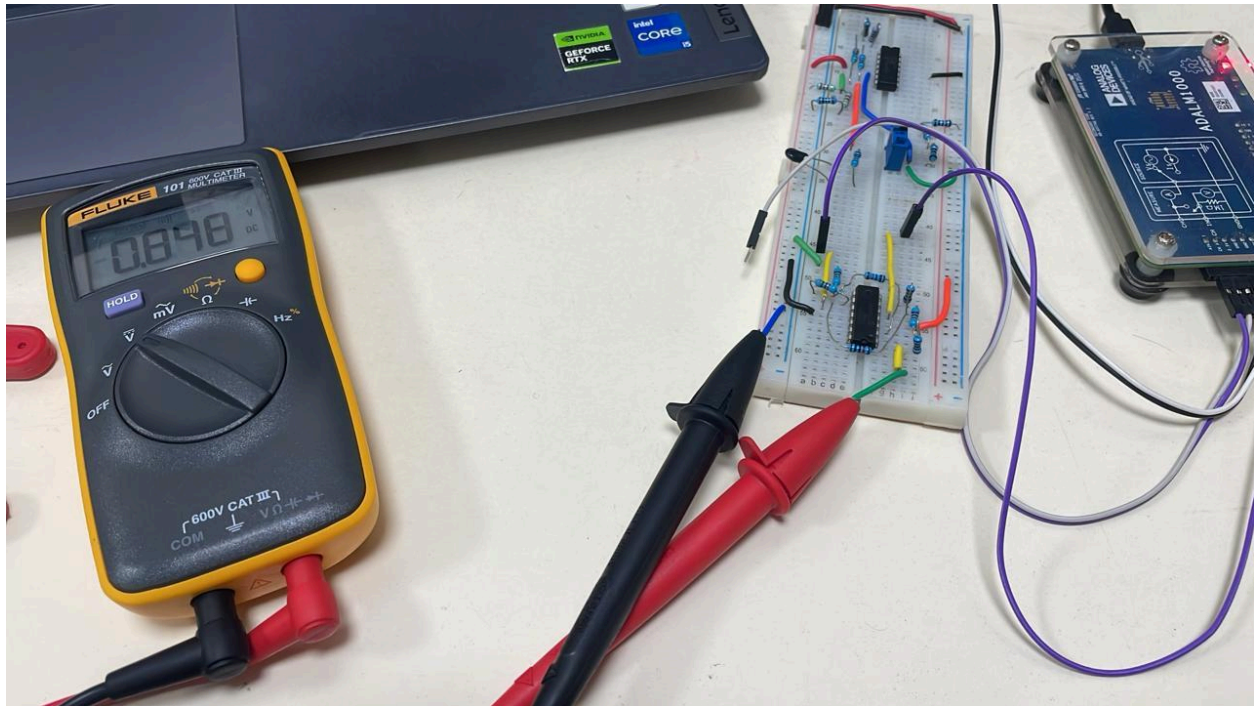
V_OUT measured using Multimeter



Full Setup For INA Analysis.



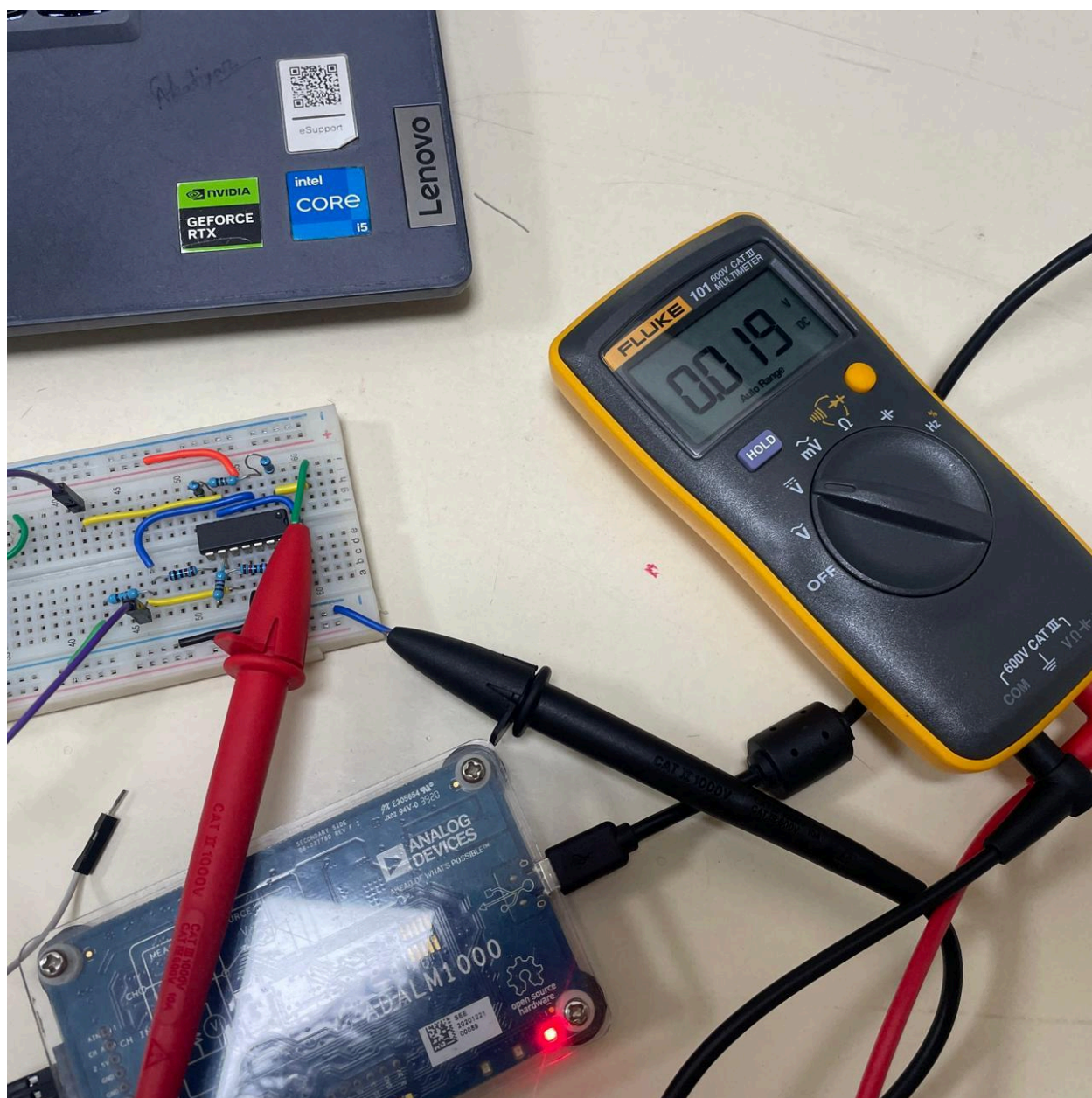
V_{out} Measurement using Multimeter



**V_{IN_2} = 1V DC fixed V_{IN_1} varied from 1V to 2V DC
in steps of 0.2V**

V_{IN_1} = 1V

Vout = 0.019V



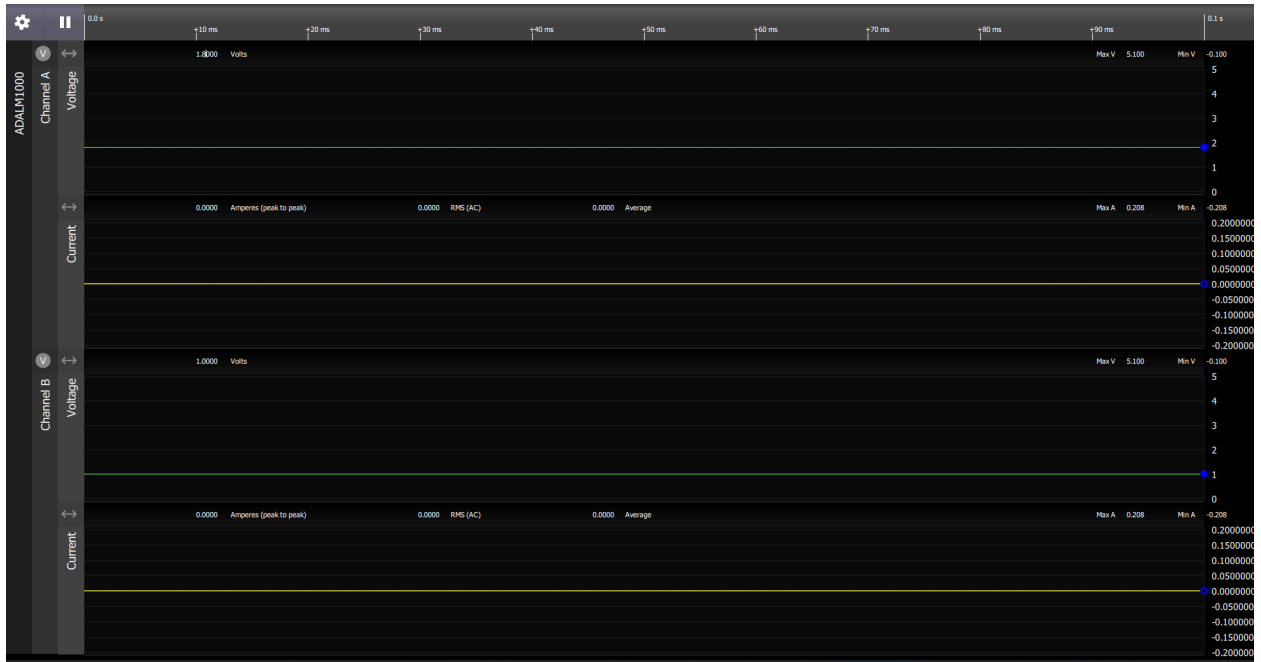
V_IN_1 = 1.4V



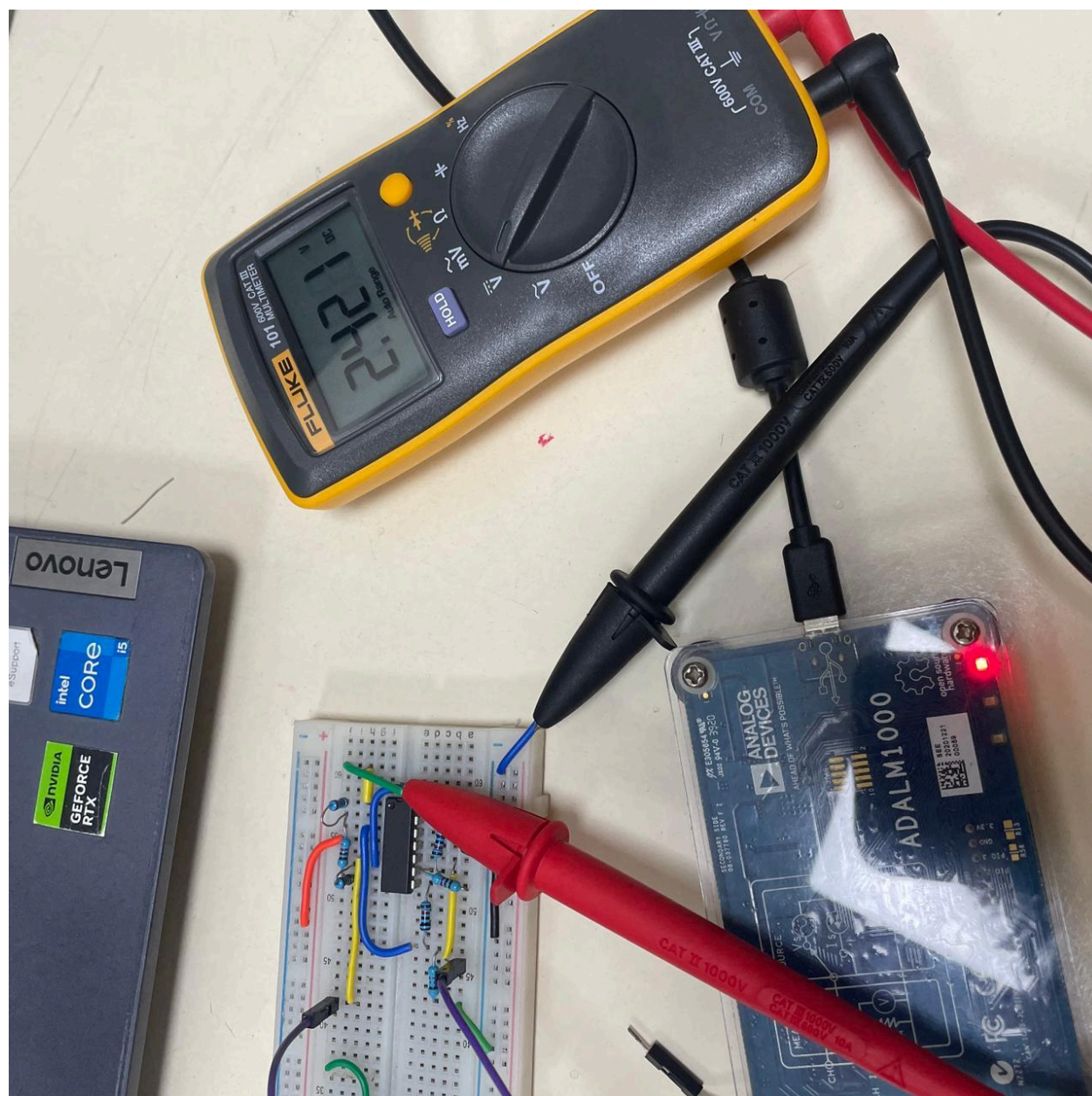
$V_{out} = 1.220V$



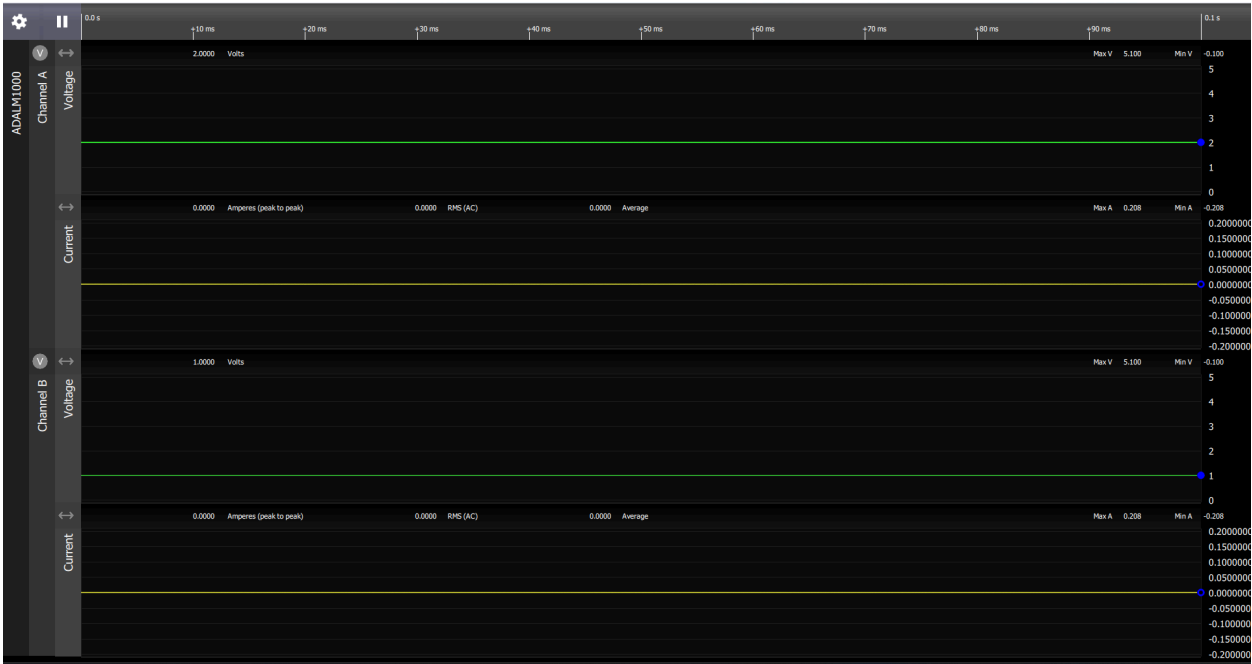
V_IN_1 = 1.8V



Vout = 2.421V



$V_{IN_1} = 2V$



$V_{out} = 3.020V$

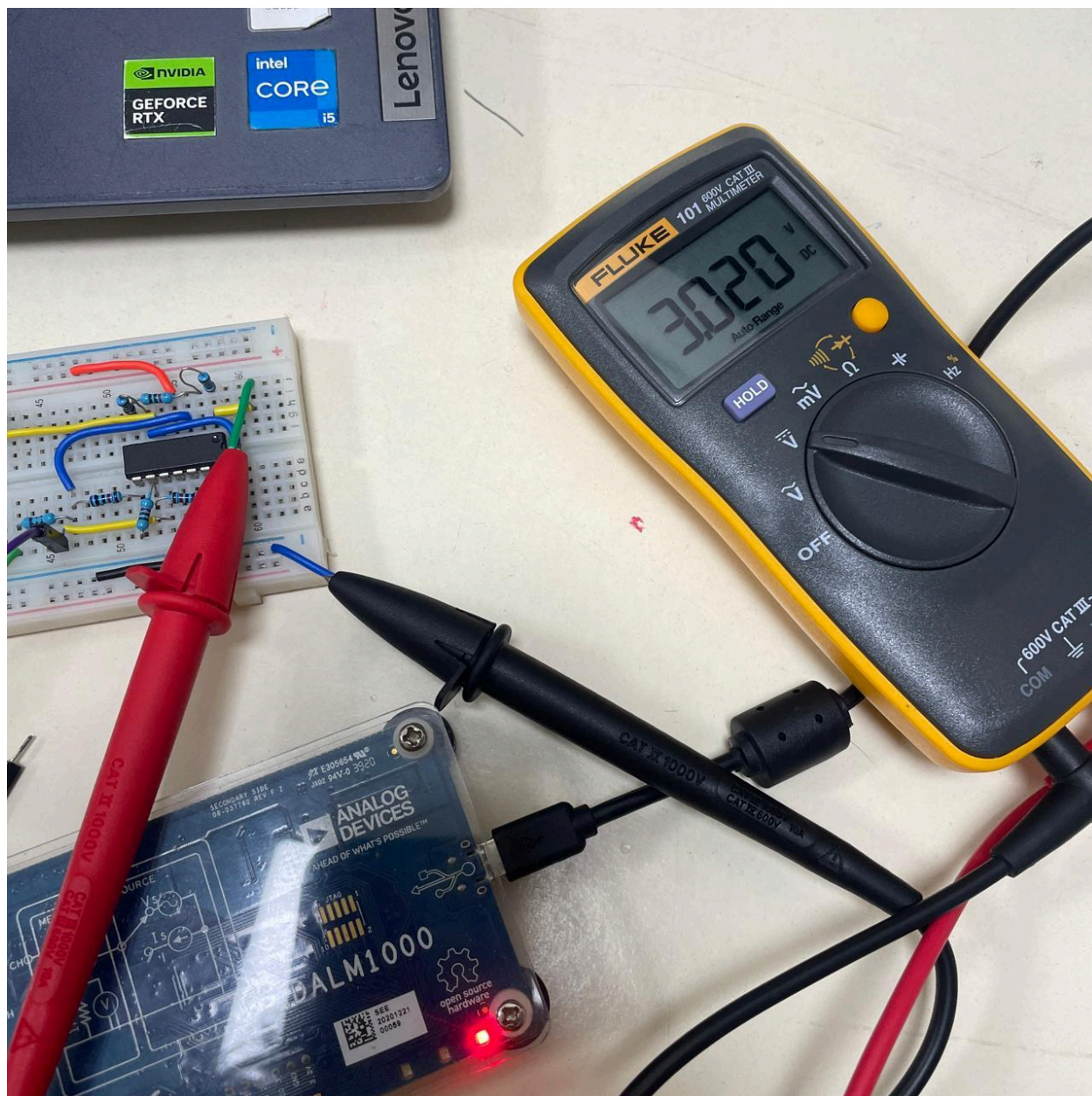
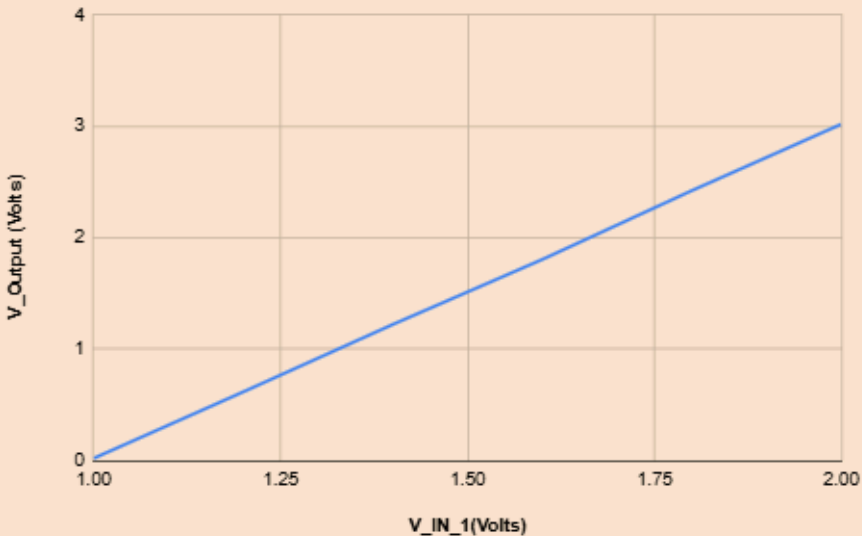


Table of Measurements for DC input

V_IN_1	V_IN_2	V_d	V_Output
1	1	0	0.019
1.2	1	0.2	0.615
1.4	1	0.4	1.21
1.6	1	0.6	1.805
1.8	1	0.8	2.399
2	1	1	2.945

DC Transfer Characteristic

V_IN_1	V_output
1	0.019
1.2	0.615
1.4	1.22
1.6	1.805
1.8	2.421
2	3.02



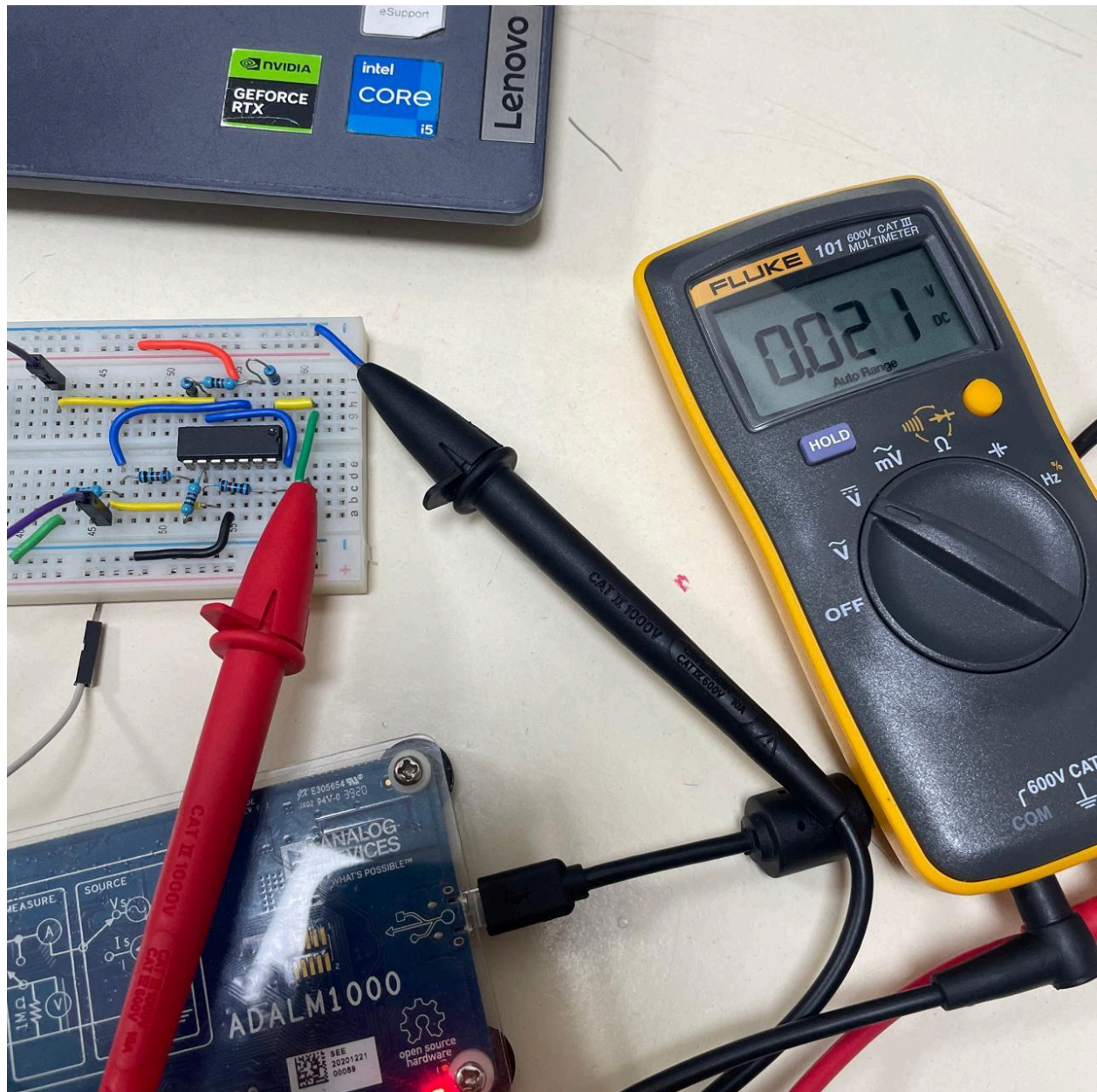
Task : 5

(Ac and CMRR)

Set $V_{IN_1} = 1.5V$ and $V_{IN_2} = 1.5V$



V_OUT is around 0.021V (or 0.020V)



$$V_{\text{out}} = A_d * V_d + A_c * V_c$$

A_d – Differential gain, A_c – common-mode gain
 V_d – differential input voltage ($V_{\text{IN}_1} - V_{\text{IN}_2}$)
 V_c – common-mode voltage : $(V_{\text{IN}_1} + V_{\text{IN}_2})/2$

$$\text{Since } V_d = 1.5 - 1.5 = 0,$$

$$V_{\text{out}} = A_c * V_c$$

$$V_{\text{out}} = 0.019\text{V}, V_c = (1.5 + 1.5)/2 = 1.5\text{V}$$

Task 5a :

$$A_c = 0.019/1.5 = 0.0127$$

Task 5b:

$$\text{CMRR} = 20 \log (A_d/A_c) = 20 \log(3/0.0127) = 47.47 \text{ dB}$$

Task : 6

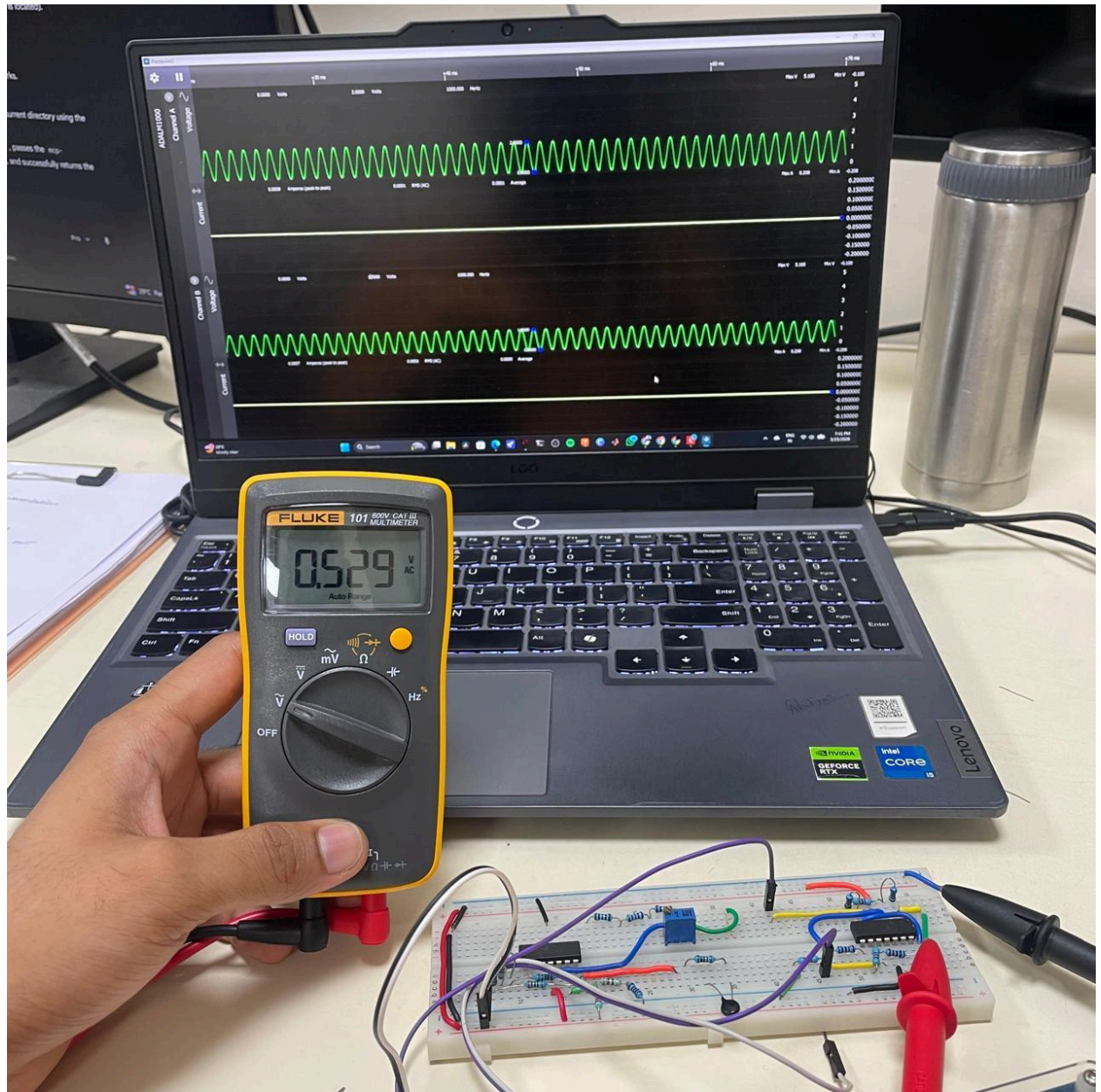
$$V_{IN_1} = 2 \sin(2\pi f t)$$
$$V_{IN_2} = 1.5 \sin(2\pi f t)$$

Frequency(Hz)	Vd	Vout	Gain (Ad)
10	0.5	0.515	1.03
100	0.5	0.532	1.064
1000	0.5	0.529	1.058
10000	0.5	0.469	0.938
20000	0.5	0.295	0.59

Frequency f = 1000 Hz (1 KHz)



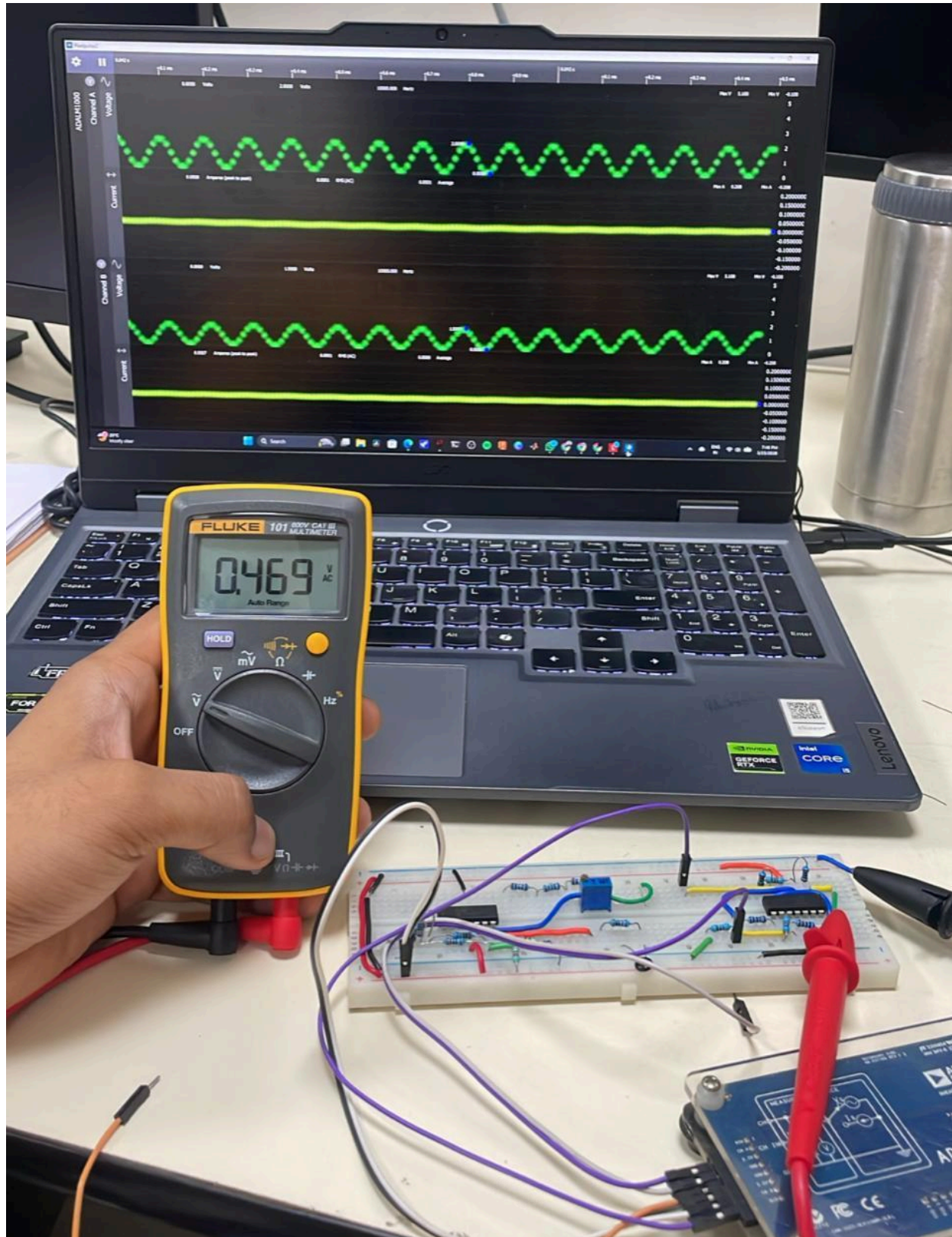
$V_{\text{out}} = 0.529 \text{ V (AC)}$



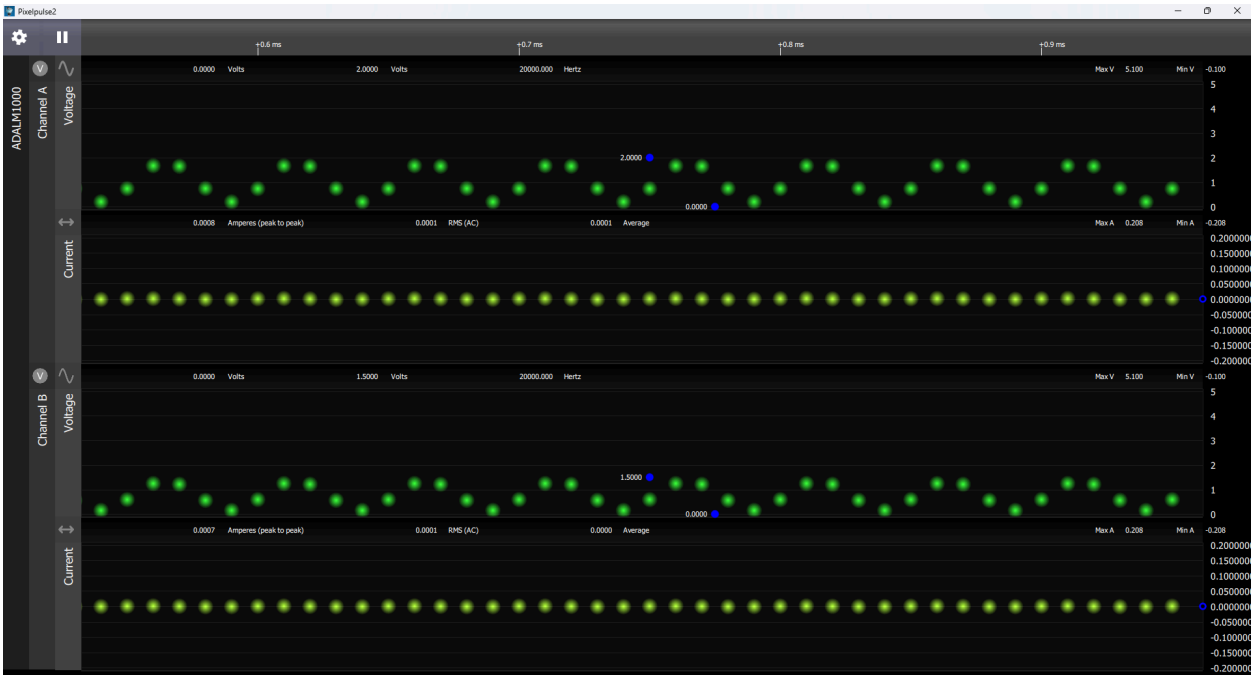
Frequency = 10 Khz



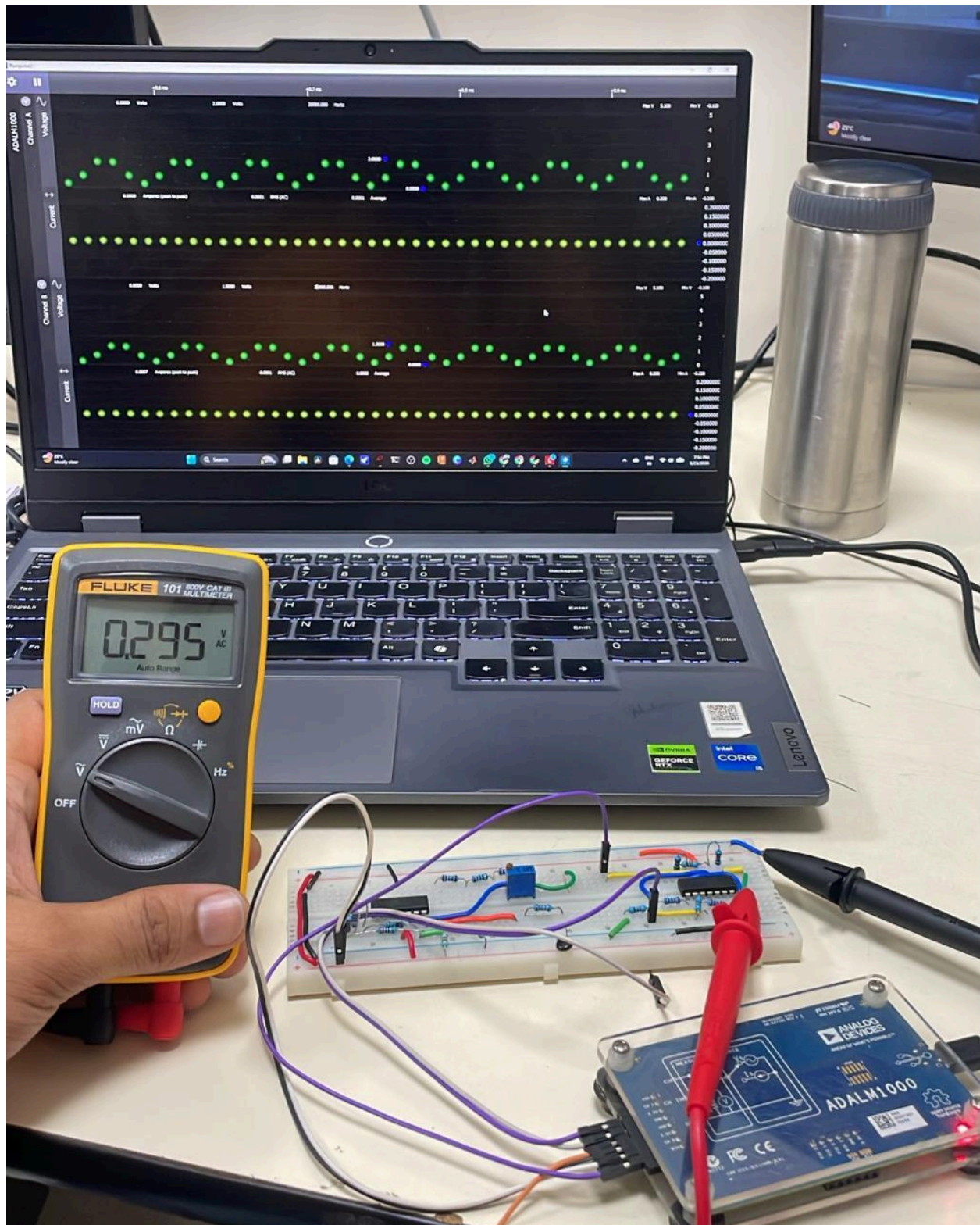
$V_{out} = 0.469 \text{ V (AC)}$



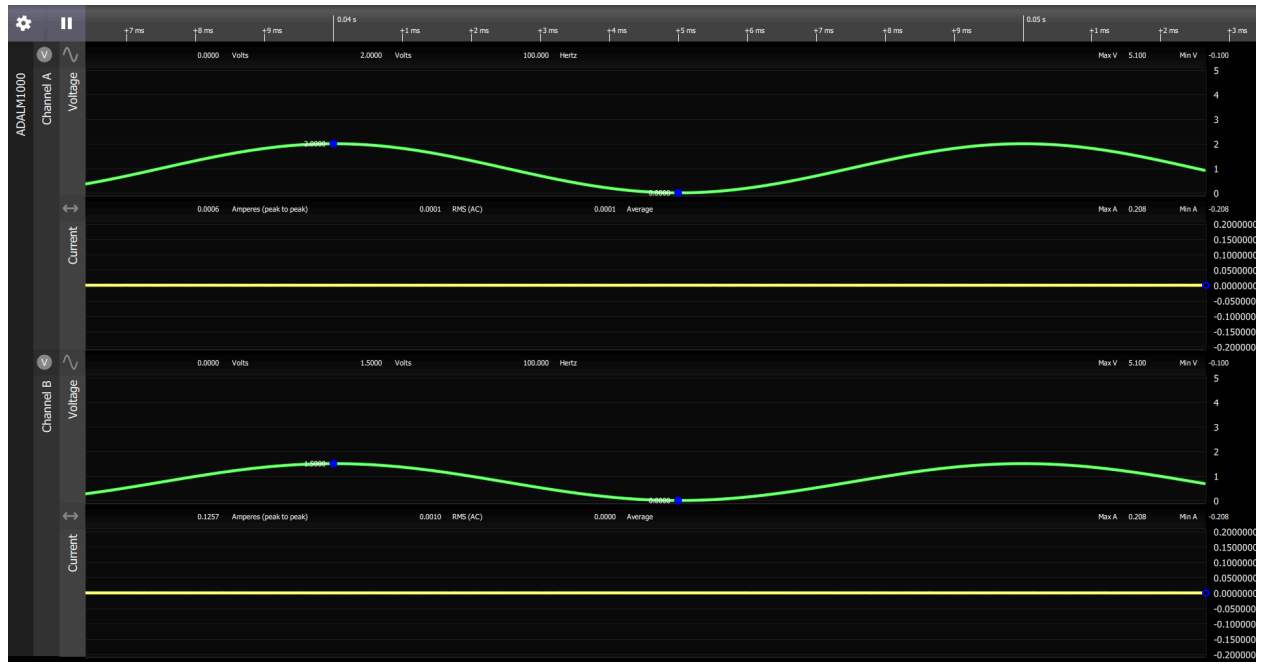
Frequency = 20 Khz



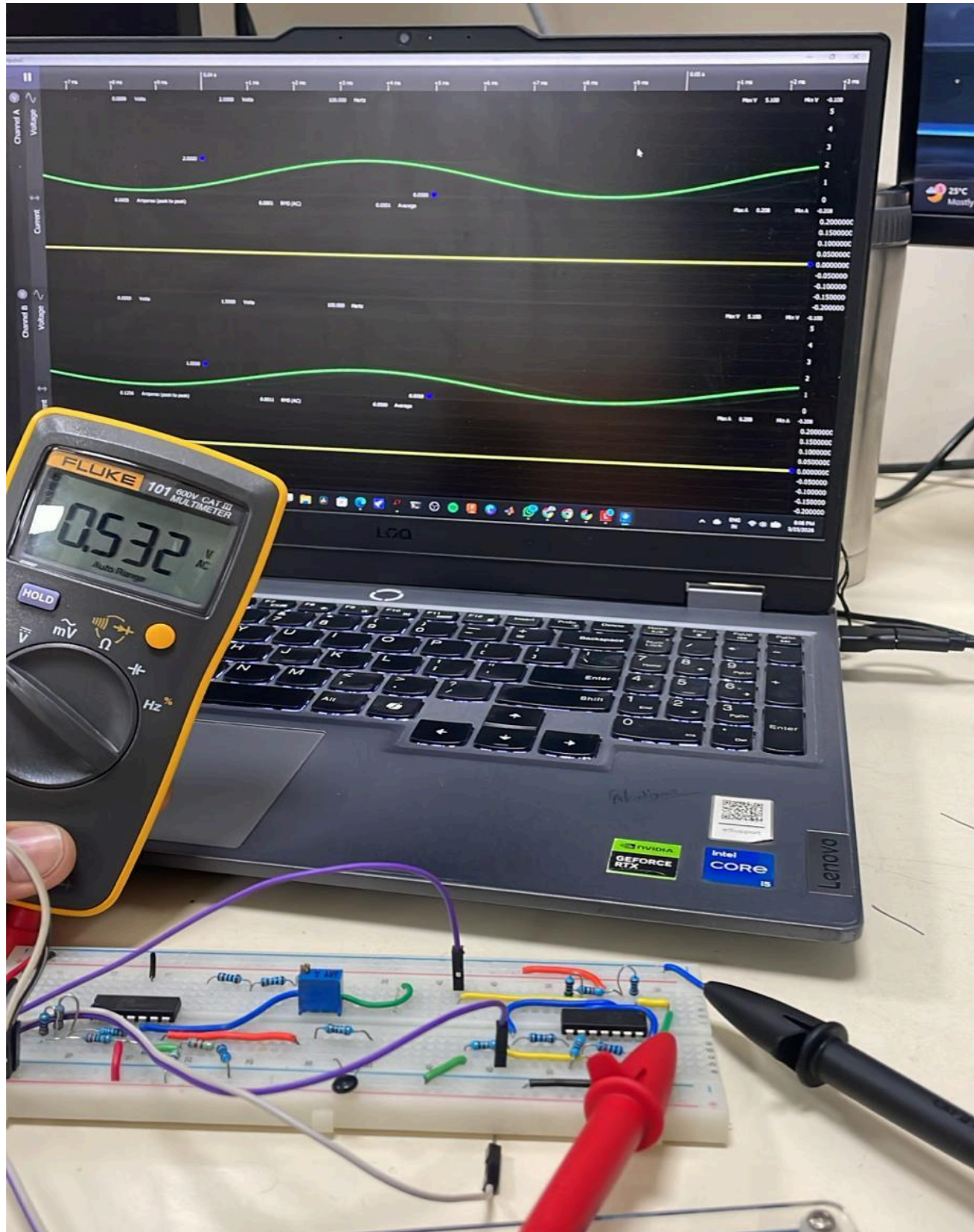
$$V_{\text{out}} = 0.295 \text{ V (AC)}$$



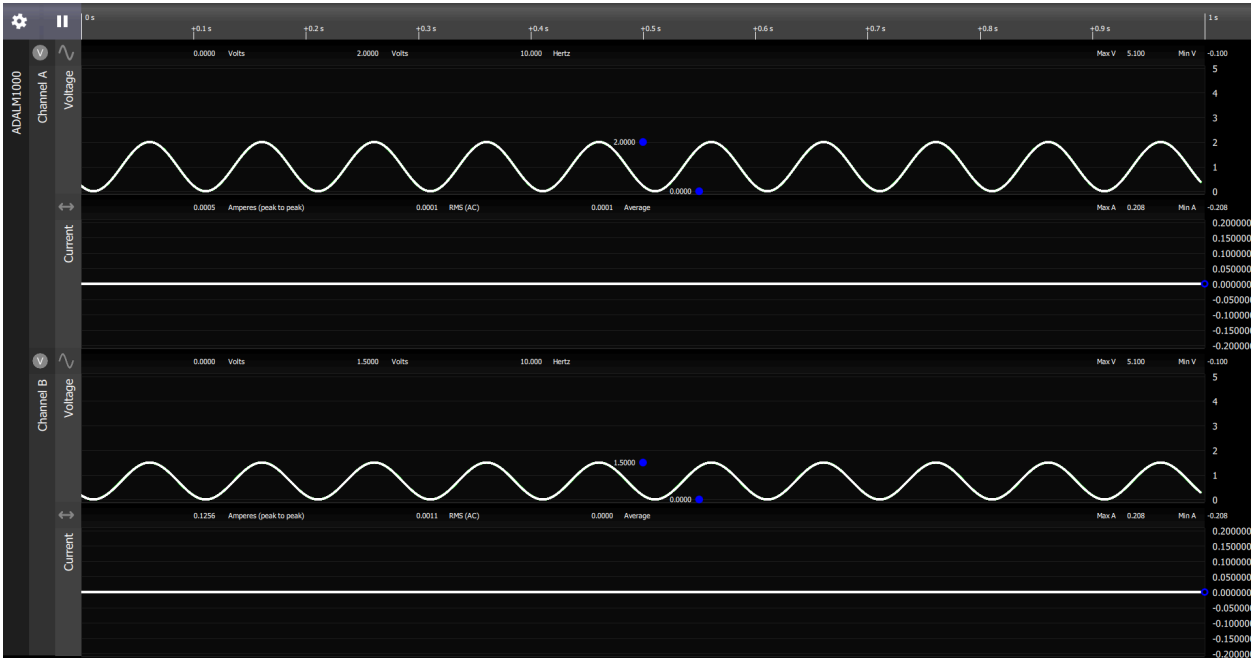
Frequency = 100 hz



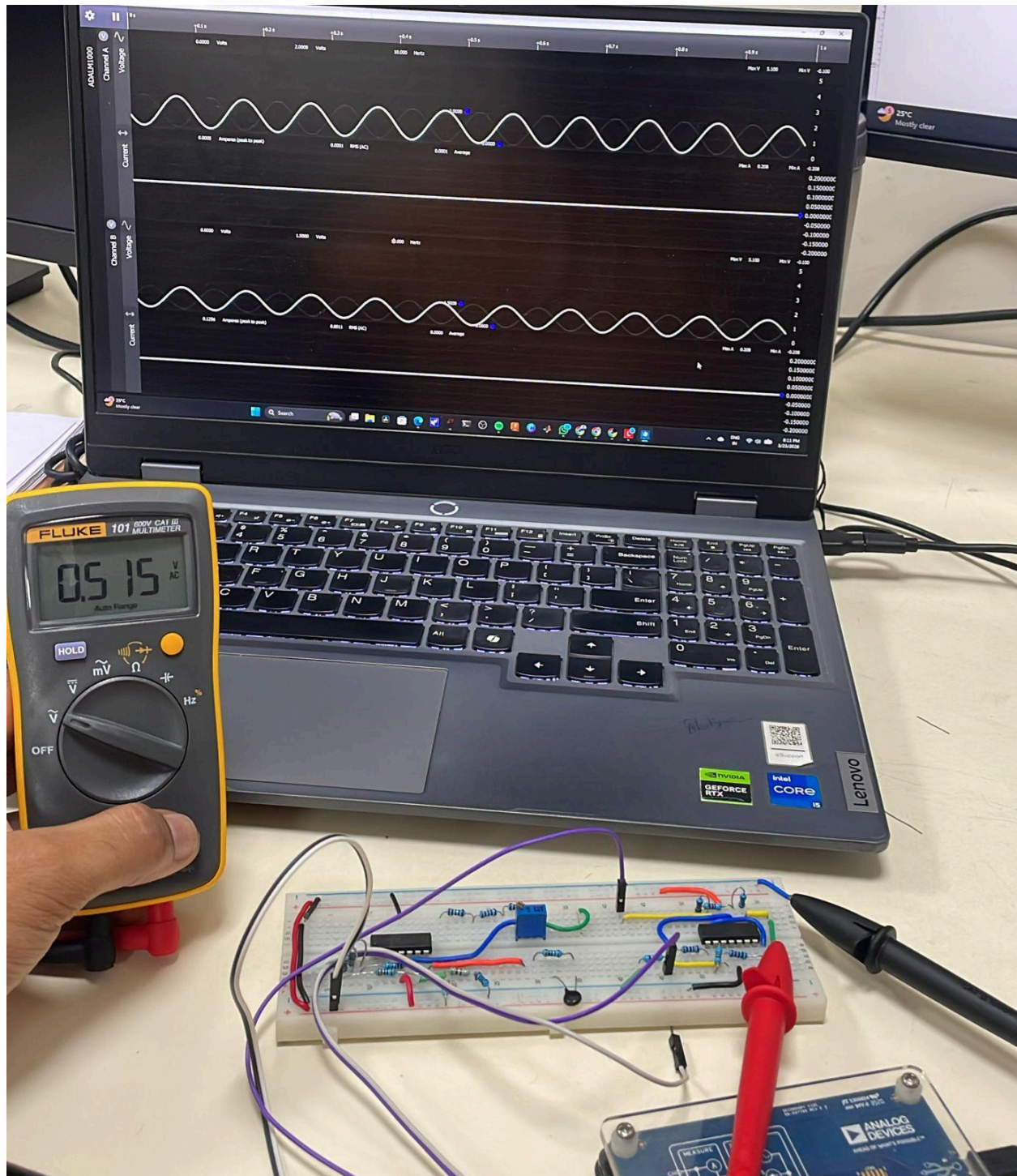
$V_{out} = 0.532 \text{ V (AC)}$



Frequency = 10 hz



$V_{out} = 0.515 \text{ V (AC)}$



Task : 7

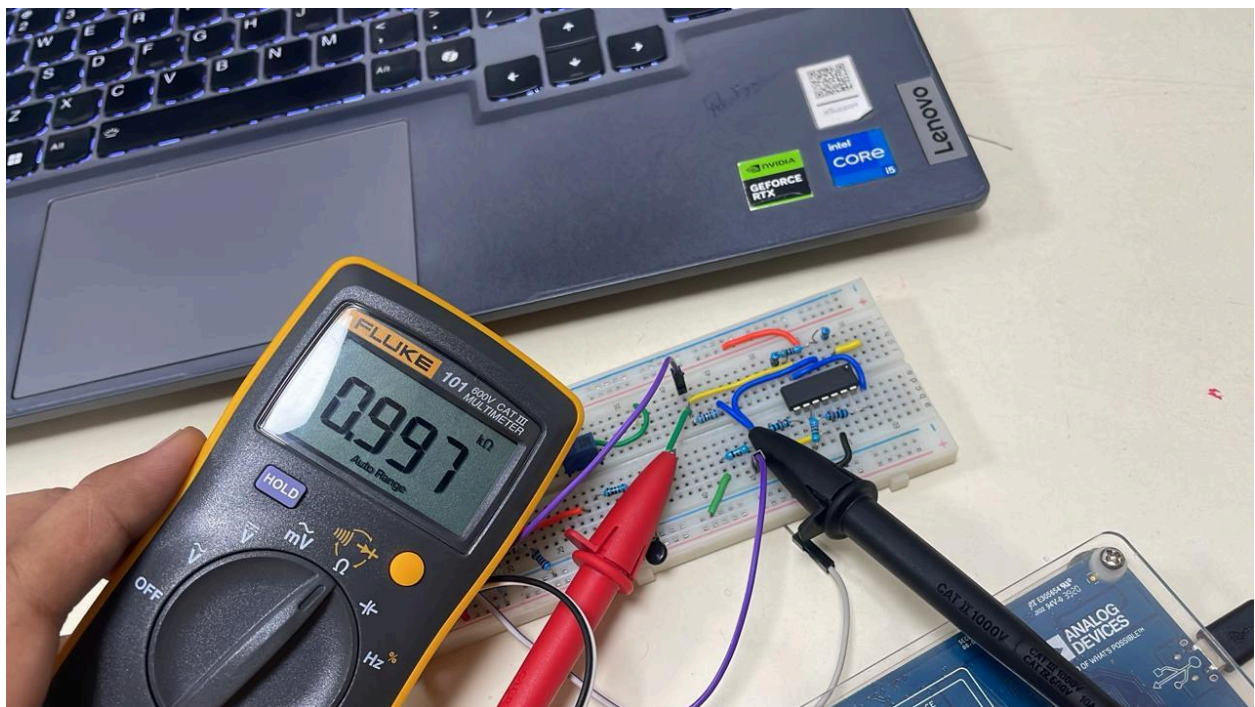
If $A_d = 20$,

$R_g = 1052.63 \text{ Ohms}$.

For the experiment, R_g is taken as 1053 Ohms

The circuit was modified like so:

Replaced R_g of 10K with two resistors in series adding up to around 1052 Ohms



Two sets of readings were taken as shown below

Units : Volts	Steps of 50 mV		
V1	V2	Vd	Vout
0.5	0.5	0	0.147
0.55	0.5	0.05	1.015
0.6	0.5	0.1	1.112
0.65	0.5	0.15	1.204
0.7	0.5	0.2	1.297
V1	V2	Vd	Vout
1	1	0	0.145
1.05	1	0.05	1.138
1.1	1	0.1	2.037
1.15	1	0.15	2.133
1.2	1	0.2	2.224

At $V_1 = V_2 = 1.5V$, $V_{out} = 0.153V$ $A_c = 0.153/1.5 = 0.102$ $CMRR = 20 \log (20/0.102) = 45.84 \text{ dB}$

Task : 8

If V_{ref} is not 0, and $R1 = R2 = R_f = R$,
 $V_{out} = (1 + 2R/R_g) * V_d + (1 + R/R) * V_{ref}/2$

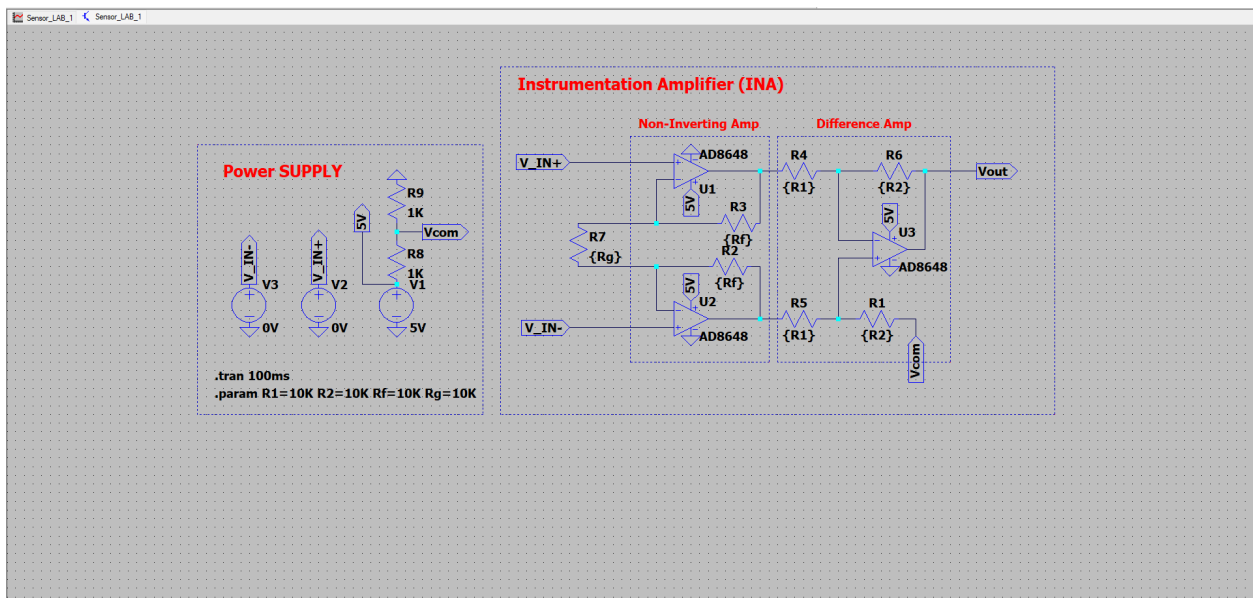
$$V_{out} = (1 + 2R/R_g) * V_d + V_{ref}$$

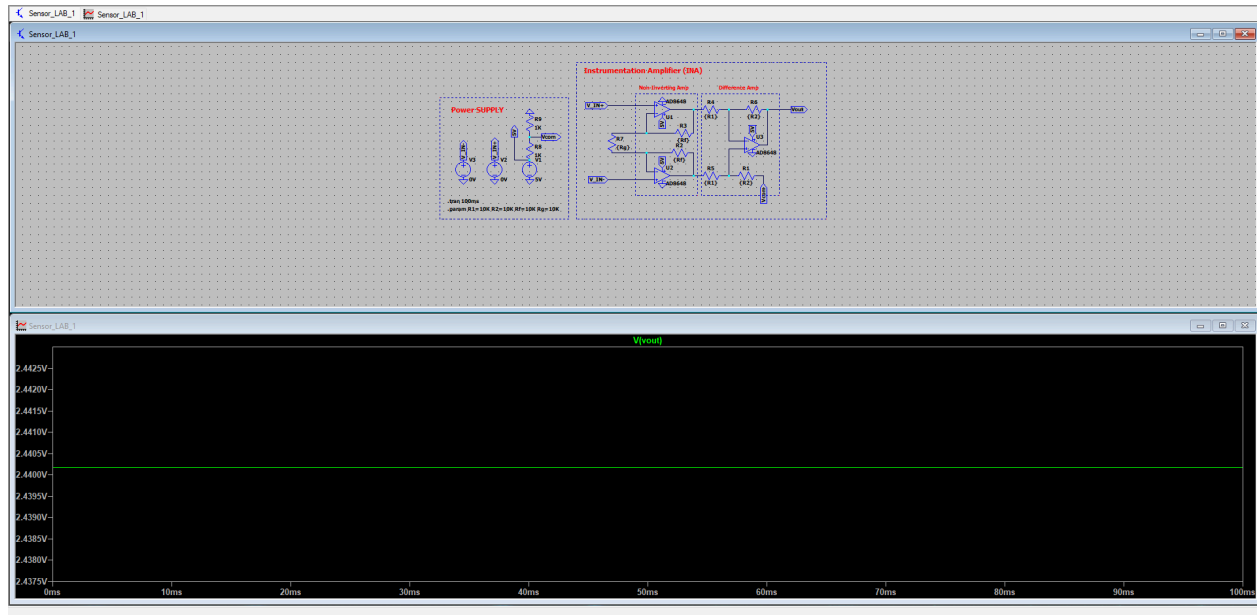
If $V_{IN_1} = V_{IN_2} = 0$, $V_d = 0$.

Therefore V_{out} should be almost equal to V_{ref} .

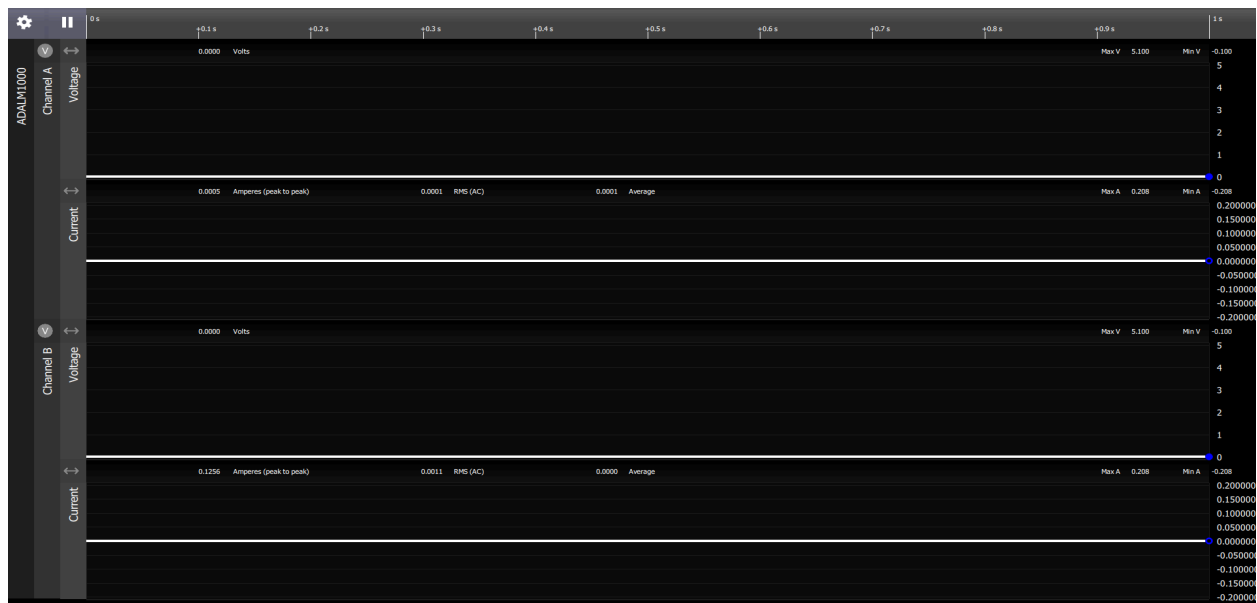
This has been verified in the simulation below.

V_{ref} is approximately 2.439V and V_{out} is seen to be approximately 2.44V

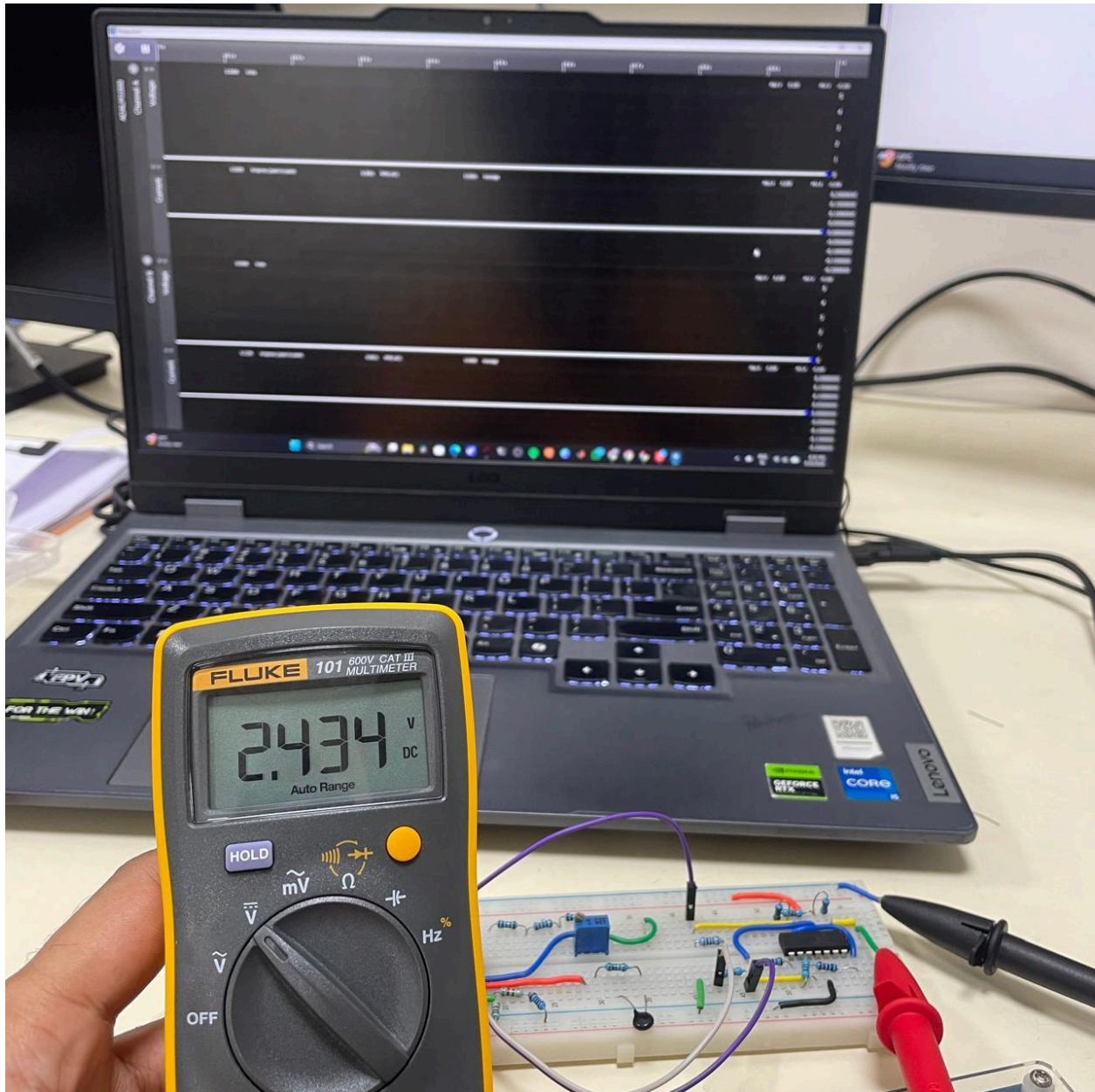




**This result has been verified in the actual circuit also:
Both inputs are 0V. Vref = 2.5V from Adalm.**



$V_{out} = 2.484V$



THANK YOU